

SINDIRI MANIKANTA SWAROOP

(+91) 88975 77277 – manikantaswaroopsindiri@gmail.com – Palasa, Andhra Pradesh

LinkedIn – GitHub – Portfolio

Experience

Summer Research Intern – CSIR-CMERI, Durgapur

May 2025 - Jul 2025

- Built and optimized CNN models for sonar-based AUV docking, achieving **95.6% classification accuracy** on a dataset of **4,000+ sonar images**.
- Built an automated data pipeline (.son → .ppm) in PyTorch, enabling **real-time inference at 20 FPS**.
- Applied DenseNet121 transfer learning, GAN-based augmentation, and hyperparameter tuning, improving model robustness by **18% under noisy conditions**.
- Collaborated with a team of **4 researchers** on sonar-based experimentation.
- Reduced experimental iteration time by **30%** across **12 validation runs**.

Technical Skills

Programming Languages	: Python, Java, SQL
Web Technologies	: HTML, CSS, JavaScript, JDBC, SpringBoot, React, Core Java
Machine Learning	: CNNs, Supervised & Unsupervised Learning, Transfer Learning
Database Technologies	: MySQL, Database Management, AWS
Tools & Libraries	: PyTorch, NumPy, Pandas, Linux, Matplotlib, Git, VsCode, IntelliJ IDEA,

Projects

Autonomous Underwater Docking using Sonar-based CNN [GitHub](#)

- Trained CNN models achieving **95.6% accuracy** for AUV docking orientation using forward-looking sonar data.
- Engineered a real-time inference pipeline in PyTorch, reducing end-to-end processing latency by **25%**.

Tiles Showroom Web Application [GitHub](#)

- Created a responsive product catalog using HTML, CSS, and JavaScript (**50+ products**).
- Integrated inquiry form submission and validation logic.
- Reduced average customer response time by **40%**.

BharatBox – IoT-Based Machine Health Monitoring System [GitHub](#)

- Engineered an ESP32-based IoT machine health monitoring system integrating temperature, vibration, and current sensors for MSME industries.
- Built a web dashboard that monitored machine health and generated fault alerts within **2 seconds**.

Achievements & Certifications

- **3rd Place – Sparktank Hackathon 2025:** Built “BharatBox,” an IoT-based machine monitoring system (20,000 award).
- **IASc-INSa-NASI Research Fellow:** Selected among the **top 1% nationwide**.
- **NPTEL Certified – Artificial Intelligence:** **81%** score (proctored national-level exam).
- Presented a technical paper at the GMRIT Intercollege Symposium.
- Top 3 finalist in the “Samukuth” Coding Competition at ANITS.

Education

Vishnu Institute of Technology, Bhimavaram

Sep 2023 - Present

- **B.Tech in Information Technology**
CGPA: **8.81 / 10.0**

Expected Graduation: **May 2027**

Leadership

- **College Fest Organizer:** Coordinated logistics and scheduling for events with **300+ participants**.
- **DoIT Club Organizer:** Conducted hands-on coding workshops attended by **60+ students**.
- Mentored junior students during hackathons and practice sessions, helping **3 teams** reach final rounds.